

Project Title
The News & Social Media Analytics Engine
Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

The News & Social Media Analytics Engine is an intelligent and scalable platform developed to collect, organize, analyse, and manage large volumes of news articles and social media content from multiple digital sources. With the rapid growth of online news portals and social networking platforms, users often encounter duplicate articles, misinformation, scattered information, and difficulty in identifying trending events. The proposed system addresses these challenges by providing a centralized platform that performs intelligent content analysis and delivers meaningful insights in real time.

The system offers a wide range of features, including keyword-based news search, advanced filtering by category, date, source, and location, duplicate content detection, related story clustering, sentiment analysis, trending topic identification, hashtag tracking, fake news detection, personalized news recommendations, moderator assignment, user management, and interactive analytics dashboards. It also supports real-time monitoring of social media discussions, engagement analysis through likes, comments, and shares, and generates statistical reports to help users understand public opinion and emerging trends. The platform is capable of processing large-scale datasets efficiently while maintaining fast response times and high accuracy.

Unlike existing systems that primarily focus on displaying news or collecting social media posts, the proposed system integrates multiple intelligent analytical features into a single platform. It automatically identifies duplicate articles to eliminate redundant information, groups related news into unified story clusters, tracks viral topics as they emerge, and assists moderators in reviewing suspicious or misleading content through automated task allocation. The recommendation module provides users with personalized content based on their interests and reading behaviour, while the analytics dashboard presents visual summaries of trending events, sentiment distribution, user engagement, and content popularity.

Signature of the Guide